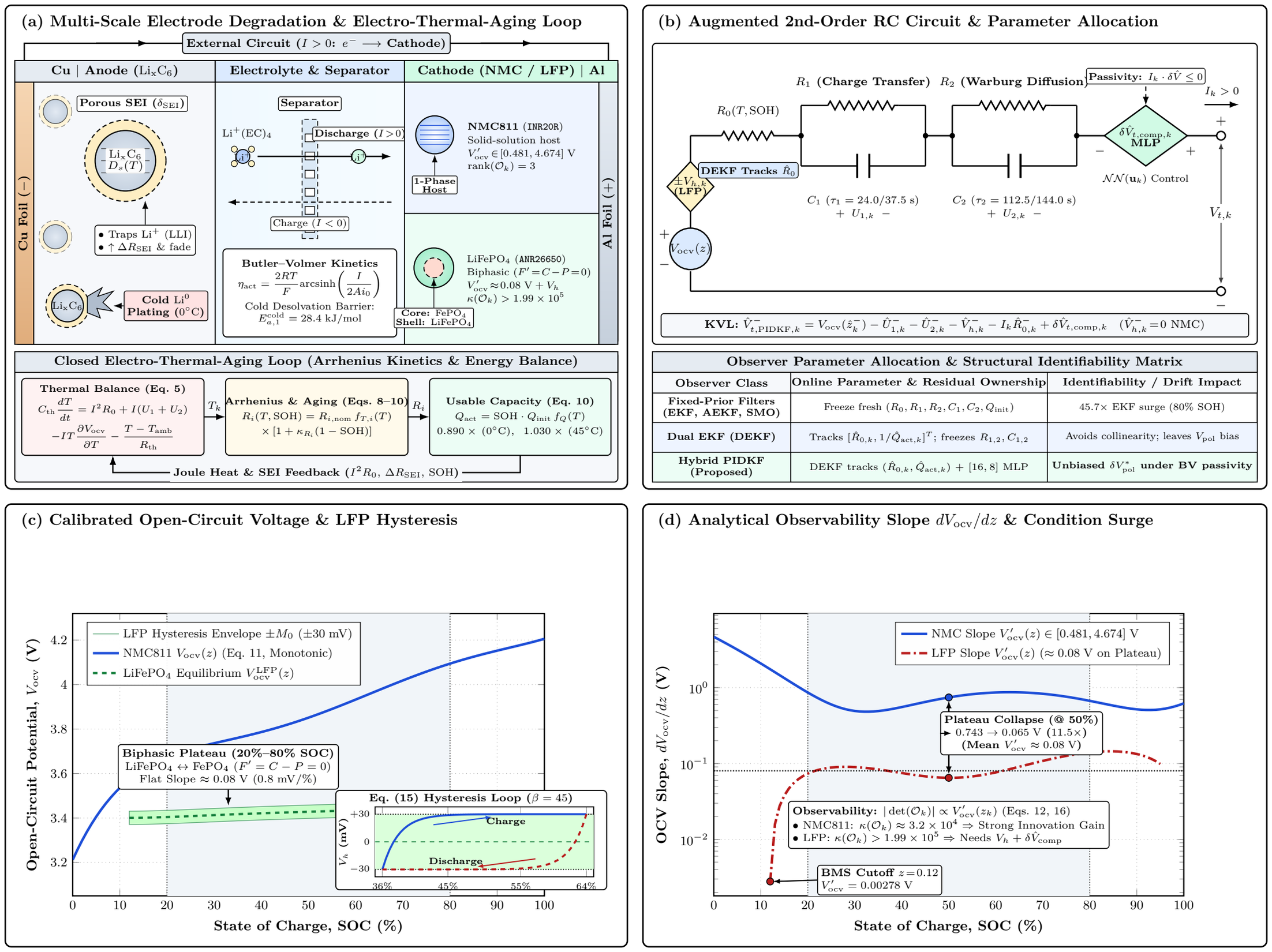

(a) Multi-Scale Electrode Degradation & Electro-Thermal-Aging Loop
External Circuit ($I > 0$: $e^- \longrightarrow$ Cathode)
Cu | Anode (Li_xC_6)
Electrolyte & Separator
Cathode (NMC / LFP) | Al
Cu Foil (−)
Al Foil (+)
Porous SEI (δ_{SEI})
Li_xC_6 $D_s(T)$
• Traps Li^+ (LLI)
• ↑ ΔR_{SEI} & fade
Li_xC_6
Cold Li^0 Plating (0°C)
Separator
$Li^+(EC)_4$
Discharge ($I > 0$)
Charge ($I < 0$)
Butler–Volmer Kinetics
$\eta_{act} = \frac{2RT}{F}\text{arcsinh}\left(\frac{I}{2Ai_0}\right)$
Cold Desolvation Barrier:
$E_{a,1}^{cold} = 28.4$ kJ/mol
NMC811 (INR20R)
Solid-solution host
$V'_{ocv} \in [0.481, 4.674]$ V
rank$(\mathcal{O}_k) = 3$
1-Phase Host
$LiFePO_4$ (ANR26650)
Biphasic ($F' = C - P = 0$)
$V'_{ocv} \approx 0.08$ V + V_h
$\kappa(\mathcal{O}_k) > 1.99 \times 10^5$
Core: $FePO_4$
Shell: $LiFePO_4$
Closed Electro-Thermal-Aging Loop (Arrhenius Kinetics & Energy Balance)
Thermal Balance (Eq. 5)
$C_{th}\frac{dT}{dt} = I^2R_0 + I(U_1 + U_2)$
$-IT\frac{\partial V_{ocv}}{\partial T} - \frac{T - T_{amb}}{R_{th}}$
T_k
Arrhenius & Aging (Eqs. 8–10)
$R_i(T, \text{SOH}) = R_{i,nom}\, f_{T,i}(T)$
$\times [1 + \kappa_{R_i}(1 - \text{SOH})]$
R_i
Usable Capacity (Eq. 10)
$Q_{act} = \text{SOH} \cdot Q_{init}\, f_Q(T)$
0.890 × (0°C), 1.030 × (45°C)
Joule Heat & SEI Feedback (I^2R_0, ΔR_{SEI}, SOH)
(b) Augmented 2nd-Order RC Circuit & Parameter Allocation
R_1 (Charge Transfer)
R_2 (Warburg Diffusion)
Passivity: $I_k \cdot \delta\hat{V} \leq 0$
$I_k > 0$
$R_0(T, \text{SOH})$
$\delta\hat{V}_{t,comp,k}$ MLP
$\mathcal{NN}(\mathbf{u}_k)$ Control
DEKF Tracks $\hat{R}_0$
$\pm V_{h,k}$ (LFP)
C_1 ($\tau_1 = 24.0/37.5$ s)
+ $U_{1,k}$ −
C_2 ($\tau_2 = 112.5/144.0$ s)
+ $U_{2,k}$ −
$V_{t,k}$
$V_{ocv}(z)$
KVL: $\hat{V}^-_{t,\text{PIDKF},k} = V_{ocv}(\hat{z}^-_k) - \hat{U}^-_{1,k} - \hat{U}^-_{2,k} - \hat{V}^-_{h,k} - I_k\hat{R}^-_{0,k} + \delta\hat{V}_{t,comp,k}$ ($\hat{V}^-_{h,k} = 0$ NMC)
Observer Parameter Allocation & Structural Identifiability Matrix
Observer Class
Online Parameter & Residual Ownership
Identifiability / Drift Impact
Fixed-Prior Filters (EKF, AEKF, SMO)
Freeze fresh ($R_0, R_1, R_2, C_1, C_2, Q_{init}$)
45.7× EKF surge (80% SOH)
Dual EKF (DEKF)
Tracks $[\hat{R}_{0,k}, 1/\hat{Q}_{act,k}]^T$; freezes $R_{1,2}, C_{1,2}$
Avoids collinearity; leaves V_{pol} bias
Hybrid PIDKF (Proposed)
DEKF tracks ($\hat{R}_{0,k}, \hat{Q}_{act,k}$) + [16, 8] MLP
Unbiased δV^*_{pol} under BV passivity
(c) Calibrated Open-Circuit Voltage & LFP Hysteresis
LFP Hysteresis Envelope $\pm M_0$ (±30 mV)
NMC811 $V_{ocv}(z)$ (Eq. 11, Monotonic)
$LiFePO_4$ Equilibrium $V^{LFP}_{ocv}(z)$
Biphasic Plateau (20%–80% SOC)
$LiFePO_4 \leftrightarrow FePO_4$ ($F' = C - P = 0$)
Flat Slope ≈ 0.08 V (0.8 mV/%)
Eq. (15) Hysteresis Loop ($\beta = 45$)
Charge
Discharge
V_h (mV)
+30
0
−30
36%
45%
55%
64%
Open-Circuit Potential, V_{ocv} (V)
State of Charge, SOC (%)
(d) Analytical Observability Slope dV_{ocv}/dz & Condition Surge
NMC Slope $V'_{ocv}(z) \in [0.481, 4.674]$ V
LFP Slope $V'_{ocv}(z)$ (≈ 0.08 V on Plateau)
Plateau Collapse (@ 50%)
0.743 → 0.065 V (11.5×)
(Mean $V'_{ocv} \approx 0.08$ V)
Observability: $|\det(\mathcal{O}_k)| \propto V'_{ocv}(z_k)$ (Eqs. 12, 16)
• NMC811: $\kappa(\mathcal{O}_k) \approx 3.2 \times 10^4 \Rightarrow$ Strong Innovation Gain
• LFP: $\kappa(\mathcal{O}_k) > 1.99 \times 10^5 \Rightarrow$ Needs $V_h + \delta\hat{V}_{comp}$
BMS Cutoff $z = 0.12$
$V'_{ocv} = 0.00278$ V
OCV Slope, dV_{ocv}/dz (V)
State of Charge, SOC (%)